C6.3.2.1.1 General

Other material properties that may be of interest for concrete elements and components include:

1. Tensile strength and modulus of elasticity, which can be derived from the compressive strength, do not warrant the damage associated with the extra coring required.
2. Ductility, toughness, and fatigue properties of concrete.
3. Carbon equivalent present in the reinforcing steel.
4. Presence of any degradation such as corrosion, bond with concrete, and chemical composition.

The effort required to determine these properties depends on the availability of accurate updated construction documents and drawings, quality and type of construction (absence of degradation), accessibility, and condition of materials. The method of analysis selected (e.g., Linear Static Procedure, Nonlinear Static Procedure) may also influence the scope of the testing.

The size of the samples and removal practices to be followed are referenced in *FEMA 274*. Generally, mechanical properties for both concrete and reinforcing steel can be established from combined core and specimen sampling at similar locations, followed by laboratory testing. Core drilling should minimize damaging the existing reinforcing steel as much as practicable.

6.3.2.1.2 Nominal or Specified Properties

Nominal material properties, or properties specified in construction documents, shall be taken as lower bound material properties. Corresponding expected material properties shall be calculated by multiplying lower-bound values by a factor taken from Table 6-4 to translate from lower bound to expected values. Alternative factors shall be permitted where justified by test data.

6.3.2.2 Component Properties

The following component properties and as-built conditions shall be established:

1. Cross-sectional dimensions of individual components and overall configuration of the structure.
2. Configuration of component connections, size of anchor bolts, thickness of connector material, anchorage and interconnection of embedments and the presence of bracing or stiffening components.
3. Modifications to components or overall configuration of the structure.
4. Current physical condition of components and connections, and the extent of any deterioration present.
5. Presence of conditions that influence building performance.

C6.3.2.2 Component Properties

Component properties may be needed to characterize building performance properly in the seismic analysis. The starting point for assessing component properties and condition should be retrieval of available construction documents. Preliminary review of these documents shall be performed to identify primary vertical- (gravity-) and lateral load-carrying elements, systems, and their critical components and connections. In the absence of a complete set of building drawings, the design professional must perform a thorough inspection of the building to identify these elements, systems and components as indicated in Section 6.3.3.

6.3.2.3 Test Methods to Quantify Material Properties

6.3.2.3.1 General

Destructive and non-destructive test methods used to obtain in-place mechanical properties of materials identified in Section 6.3.2.1 and component properties identified in Section 6.3.2.2 shall comply with the requirements of this section. Samples of concrete and reinforcing and connector steel shall be examined for physical condition as specified in Section 6.3.3.2.